

## SIMATS ENGINEERING

### ASSESSMENT TOOL 2: Scenario-Based Questions

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0515

#### COURSE OUTCOME COVERED

**CO2:** Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks 50

Duration :60 Minutes

Annexure A

Scenario-Based Questions

#### Code of Conduct:

I **JAGAN R (192321102)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

#### Signature of the student:

#### RUBRICS FOR CALCULATION:

| Criteria                     | Weightage | Level 4 (Exemplary)                                                       | Level 3 (Proficient)                               | Level 2 (Developing)                              | Level 1 (Beginning)                               | Marks (100) |
|------------------------------|-----------|---------------------------------------------------------------------------|----------------------------------------------------|---------------------------------------------------|---------------------------------------------------|-------------|
| Understanding of Scenario    | 20%       | Demonstrates clear and complete understanding of the scenario and context | Shows good understanding with minor gaps           | Partial understanding; misses key aspects         | Misinterprets or fails to understand the scenario |             |
| Identification of Key Issues | 20%       | Accurately identifies all relevant issues and factors involved            | Identifies most key issues with minor omissions    | Identifies limited issues; some irrelevant points | Unable to identify key issues                     |             |
| Application of Concepts      | 25%       | Applies appropriate concepts effectively to address the scenario          | Applies concepts with minor errors                 | Limited or partially correct application          | Unable to apply relevant concepts                 |             |
| Decision Making              | 20%       | Makes well-reasoned, logical decisions with strong rationale              | Makes reasonable decisions with some justification | Makes weak decisions with limited justification   | No clear decisions made or rationale provided     |             |
| Clarity of Response          | 15%       | Response is clear, structured, and logically presented                    | Generally clear with minor issues                  | Somewhat unclear or poorly structured             | Disorganized and difficult to understand          |             |

## Scenario-Based Questions

### 1. Object Boundary Detection

In a medical image, boundaries between tissues need to be clearly identified.

- (a) Interpret the scenario and explain the importance of boundary detection.
- (b) Identify key challenges in detecting boundaries.
- (c) Apply a suitable edge detection method.
- (d) Justify your selection with logical reasoning.
- (e) Present your explanation clearly and systematically.

### 2. Simple Segmentation

An image contains objects clearly distinguishable from the background based on intensity.

- (a) Interpret the scenario and explain segmentation requirements.
- (b) Identify key features enabling separation.
- (c) Apply a suitable segmentation technique.
- (d) Justify your decision logically.
- (e) Present your response clearly.

### 3. Complex Segmentation

An image contains multiple regions with overlapping intensity values.

- (a) Interpret the scenario and explain segmentation challenges.
- (b) Identify key issues such as intensity overlap.
- (c) Apply an appropriate advanced segmentation method.
- (d) Justify how the method handles complexity.
- (e) Provide a structured and clear explanation.

### 4. Broken Edges

After applying edge detection, object boundaries appear discontinuous.

- (a) Interpret the scenario and explain why edges are broken.
- (b) Identify key issues affecting boundary continuity.
- (c) Apply a suitable method to improve edge connectivity.
- (d) Justify your approach logically.
- (e) Present your explanation clearly.

### 5. Combined Requirement

An industrial inspection system requires noise removal while preserving important edges for defect detection.

- (a) Interpret the scenario and define the combined requirement.
- (b) Identify key issues involving noise and edge preservation.
- (c) Apply an appropriate sequence of techniques.
- (d) Justify the order and selection with strong reasoning.
- (e) Present your response in a clear and structured format.

## **ANSWERS:**

### **1. (a) Interpret the scenario**

In a medical image, different tissues must be separated accurately. Boundary detection helps identify the exact borders of organs or tissues, enabling correct diagnosis and treatment planning.

#### **(b) Key challenges**

- Presence of image noise
- Low contrast between tissues
- Weak or blurred edges
- Uneven illumination

#### **(c) Suitable edge detection method**

Canny Edge Detection

#### **(d) Justification**

- Canny Edge Detection first smooths the image using a Gaussian filter to reduce noise. It then computes gradients, applies non-maximum suppression to obtain thin edges, and uses double thresholding with edge tracking to preserve true edges while eliminating false ones. Therefore, it provides accurate and continuous tissue boundaries.

#### **(e) Conclusion**

Canny Edge Detection is the most suitable method because it offers high accuracy, good noise immunity, and continuous edge detection, which are essential for medical images.

### **2. (a) Interpret the scenario**

The objects have intensity values that are clearly different from the background. Therefore, segmentation can be performed by separating pixels based on their intensity.

#### **(b) Key features enabling separation**

- High intensity difference
- Uniform background
- Distinct object boundaries
- Minimal overlap of pixel values

#### **(c) Suitable segmentation technique**

Thresholding (Global Thresholding)

#### **(d) Justification**

Thresholding divides the image into foreground and background using a single threshold value. Pixels above the threshold belong to the object, while those below belong to the background. Since the intensity difference is clear, thresholding is simple, fast, and highly effective.

#### **(e) Conclusion**

Global Thresholding is the best choice because it efficiently separates objects from the background when intensity values are clearly distinguishable.

### **3. (a) Interpret the scenario**

The image contains multiple objects with similar or overlapping intensity values, making it difficult to separate them using simple thresholding.

#### **(b) Key challenges**

- Overlapping intensity values
- Touching or connected objects
- Image noise
- Irregular object shapes

- Complex boundaries

**(c) Suitable advanced segmentation method**

Watershed Segmentation

**(d) Justification**

Watershed Segmentation treats the image as a topographic surface where pixel intensities represent elevations. It separates regions by identifying watershed lines between adjacent objects, making it effective even when objects touch or have overlapping intensity values.

**(e) Conclusion**

Watershed Segmentation effectively handles complex images by separating overlapping regions and producing accurate segmented objects.

**4. (a) Interpret the scenario**

After edge detection, the detected boundaries contain gaps and are not continuous, making object identification difficult.

**(b) Key issues**

- Image noise
- Weak edge responses
- Low image contrast
- Gaps between detected edge pixels

**(c) Suitable improvement method**

Morphological Closing (Dilation followed by Erosion)

**(d) Justification**

Morphological Closing first dilates the image to connect nearby edge segments and then erodes it to restore the original object size while keeping the connected boundaries. This produces continuous object edges.

**(e) Conclusion**

Morphological Closing improves boundary continuity by connecting broken edges without significantly changing the object shape.

**5. (a) Interpret the scenario**

The industrial inspection system must remove image noise while preserving important edges so that defects can be detected accurately.

**(b) Key issues**

- Random image noise
- Loss of edge information
- False edge detection
- Need for accurate defect identification

**(c) Appropriate sequence of techniques**

- Median Filtering – Removes impulse (salt-and-pepper) noise while preserving edges.
- Canny Edge Detection – Detects strong and weak edges accurately.

**(d) Justification**

Median Filtering is applied first because it removes noise without blurring important edges. The cleaned image is then processed using Canny Edge Detection, which accurately detects defect boundaries with minimal false edges. Applying edge detection before noise removal would increase false detections.

**(e) Conclusion**

The sequence Median Filtering → Canny Edge Detection provides effective noise reduction while preserving important edges, making it ideal for industrial defect detection.

